

UGTS-KC 4.1.1

ATLAS SPATIAL SEED SLAM FUSION

POCO X7 Pro 12 GB Android merge and repair

Capture	Camera2 YUV + Android IMU
SLAM	FAST/BRIEF, matching, visual motion, guarded triangulation, semi-dense fusion
Authority	Eight-gate proposals, deterministic ledger, stable IDs, pre/post hashes
Storage	KSEED 4.1 seed plus measured evidence deltas
Native	arm64-v8a C++ seed/CRC module with Java oracle/fallback
Status	Host/source validated; Android assemble and physical POCO run pending

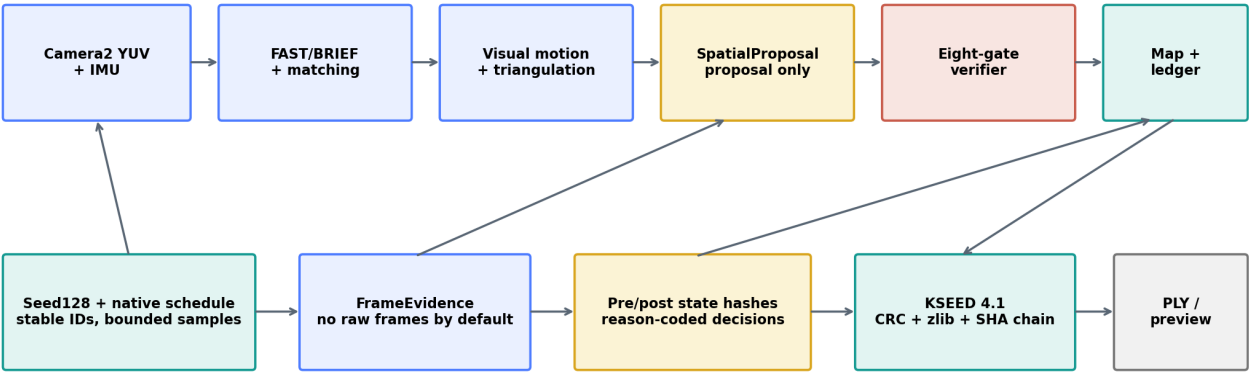
Prepared 28 August 2026

Release type: complete source handoff; no newly built APK or physical-phone benchmark claimed

1. Executive merge decision

DECISION - RELEASE AS 4.1.1. Preserve the repaired Camera2/SLAM application, add the documented Spatial Seed/KSEED/native authority layer, and keep every observation proposal-only until the ordered verifier accepts it.

The delivered application is a hybrid native Android scanner: the available 3.9.4.1 Java Camera2 and SLAM source remains the active front end, while a new arm64 C++/JNI module supplies deterministic seed scheduling and CRC acceleration. The platform-independent Java core remains the behavioral oracle and map authority. This reduces integration risk while importing the most important 4.1 mechanisms: seed/evidence separation, stable IDs, proposal gates, state hashes and KSEED storage.

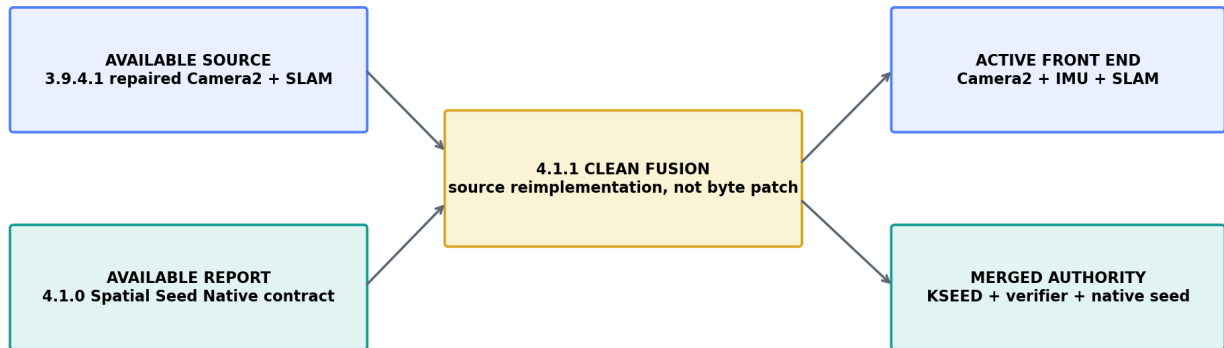


Only the verifier can turn an observation into an authoritative map mutation.

Figure 1. Merged capture, authority and storage path.

Priority	4.1.1 implementation result
Accuracy	Camera intrinsics where available, IMU rotation compensation, guarded matches/triangulation, metric anchor boundary
Efficiency	Single latest-frame queue, bounded features/voxels/keyframes, adjacent-keyframe heavy data only, arm64-only native m
Compression	No raw frames by default; delta/varint/Morton records; zlib only when smaller; exact stored-size summary
Integrity	Header/chunk CRC32, SHA-256 chain, deterministic proposal and pre/post state hashes
File size	No AndroidX, OpenCV, network stack or bundled model weights; R8/resource shrinking retained

2. Source lineage and non-negotiable merge boundary



The unavailable 4.1.0 source ZIP is not silently claimed as merged code.

Figure 2. Available implementation source and available 4.1 specification are visibly separated.

SOURCE BOUNDARY. The complete UGTS-KC 4.1.0 source ZIP was not present in the active workspace. This release therefore reimplements the documented 4.1 KSEED and authority contract against the available repaired 3.9.4.1 source. It is not a byte-identical patch of an unavailable tree.

Source	Available form	Treatment
UGTS-KC 3.9.4.1 repaired Android	ZIP source	Direct extracted baseline; Gradle repair, Camera2, IMU, SLAM, export and host tooling retained
UGTS-KC 4.1.0 Spatial Seed Native	Technical PDF	KSEED framing, verifier order, seed/evidence boundary, synthetic isolation, native profile and pro
KC Atlas 3.9.3	PDF + earlier package lineage	Observation versus authority, uncertainty, stable IDs, replay and Android bridge discipline retained
KC Two Hands 3.0 / SCLP / UGTS-KC	Object PDFs	Proposal/commit, finite packing, precision and compression boundaries retained

No global mechanism catalog was renumbered in this repair. The source report and available packages contain overlapping historical catalog ranges; 4.1.1 records implementation capabilities and contracts without silently rewriting those catalogs.

3. Camera2, IMU and SLAM front end

The active front end remains focused on the requested scanner rather than replacing it with a storage demonstration. Camera2 acquires YUV_420_888 through an ImageReader with maxImages=2 and acquireLatestImage backpressure. Luma is resampled directly; no RGB conversion is required for feature work. Android orientation and linear-acceleration samples are timestamped against camera frames.

Stage	Implemented behavior	Bound / limitation
Features	Deterministic FAST-9 and fixed 256-bit BRIEF	1,100 feature default; texture-dependent
Matching	Compact LSH candidates, mutual/ratio guards	Repetitive texture can remain ambiguous
Motion	IMU-rotation-compensated epipolar direction	Translation scale is relative before anchor
Sparse geometry	Positive-depth, ray-gap, parallax and reprojection-guarded triangulation	No global bundle adjustment
Semi-dense	Adjacent-keyframe plane sweep and confidence-weighted voxel fusion	Not learned dense depth; bounded work
Loop closure	Signature candidate recorded	Commit deferred until geometric bundle adjustment exists
Scale	Known-distance camera displacement anchor	Anchor sets scale; it does not remove drift

The app keeps the higher-detail camera preview as a downstream aid because scanning accuracy matters. The Bayer path is optional and never replaces measured luma or map authority.

```
Camera/IMU -> measured frame evidence -> tracking proposal
           -> eight-gate verification -> accepted keyframe/map mutation
```

4. Spatial seed, stable identity and reconstructibility

Every session carries a 128-bit Seed128. It deterministically schedules bounded pixel samples, creates stable proposal/keyframe/session identifiers and recreates the synthetic fixture. The seed is stored in the KSEED header. The narrow C++ module implements the same SplitMix64 schedule and CRC32 behavior as the Java oracle.

The seed does not reconstruct unstored photons. Real luma statistics, IMU summaries, accepted map evidence, uncertainty and event data remain measured records. The seed is not encryption.

Object	Identity rule	Persistence
Session	Seed + session ID + start time	KSEED header
Proposal	Seed-derived stable proposal ID	Ledger decision
Keyframe	Seed-derived stable keyframe ID; integer sequence retained	Keyframe chunk
Voxel	Signed 21-bit x/y/z cell encoded into exact Morton key	Sorted key delta
State	SHA-256 chain over prior state, canonical proposal and committed hashes	Proposed

Coordinates remain measurements, not identity. A stable node may accumulate new evidence or move after an accepted event without being silently replaced by coordinate coincidence.

5. Proposal verification and ledger authority

Ordered proposal verification

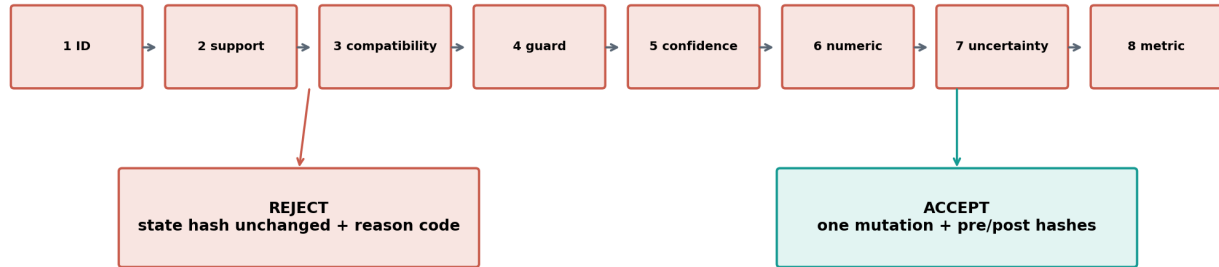


Figure 3. The verifier evaluates all required gates in a fixed order.

Gate	Reject reason	Purpose
Identifier	identifier_invalid	Stable and schema-safe proposal/entity IDs
Support	outside_support	Observation lies inside admitted local evidence support
Compatibility	incompatible	Camera/state/mode/topology policy agrees
Guard class	guard_tangency / coincident / unknown	Only predeclared classes commit
Confidence	confidence_below_floor	Measured confidence clears policy
Numeric margin	numeric_error_exceeds_margin	Finite numerical error stays within event margin
Uncertainty	uncertainty_exceeds_policy	Uncertainty remains below application maximum
Metric	metric_unavailable	Metric claims require an accepted metric anchor

Rejected proposals retain their canonical proposal hash and reason but leave the authoritative state hash unchanged. Accepted proposals receive one ordered mutation, a sequence number and distinct pre/post hashes. Camera, IMU, synthetic and future model outputs use the same record shape.

6. KSEED 4.1 stream format

128-byte session header

magic/version	flags	seed 128	start time	size/fps/budget	profile SHA-256	calibration SHA-256	CRC32
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64-byte chunk header + stored payload

type/flags	sequence/count	decoded/stored length	decoded/stored CRC32	schema	SHA-256 chain	stored payload
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Chunk order



$$H_i = \text{SHA256}(H_{(i-1)} \parallel \text{chunk_header}[0:32] \parallel \text{stored_payload})$$

Figure 4. Reconstructed 4.1 framing implemented by the Java writer and independent readers.

The 128-byte header declares version, storage mode, seed, start time, analysis dimensions, requested frame rate, feature budget and two SHA-256 profile commitments. CRC32 covers header bytes 0-123.

Each 64-byte chunk header declares type, flags, sequence, record count, decoded/stored lengths, decoded/stored CRC32 values and schema ID in its first 32 bytes. The last 32 bytes carry the chained hash. Replay stops at the first framing, sequence, CRC, zlib, chain or summary failure.

```

H_i = SHA256(H_{(i-1)} || C_i[0:32] || stored_payload)
H_{-1} = SHA256("KSEED41-CHAIN")

```

Chunk	Records
Frame evidence	Delta frame/time, dimensions, luma/gradient summaries, feature/match/inlier counts, int16 quaternion and inertial hints
Keyframes	Delta ID/time, stable ID, quantized pose, signature
Ledger	Decision records, canonical proposal hash, pre/post hashes, reason and fields
Voxels	Unsigned-sorted Morton-key deltas, intensity, confidence and observation count
Calibration	Camera source, calibrated flag, scale state, metric scale, voxel size and native status
Summary	Frames/keyframes/events/voxels/raw bytes/stored bytes plus rejection, state and chunk counts

7. Compression, efficiency and file-size discipline

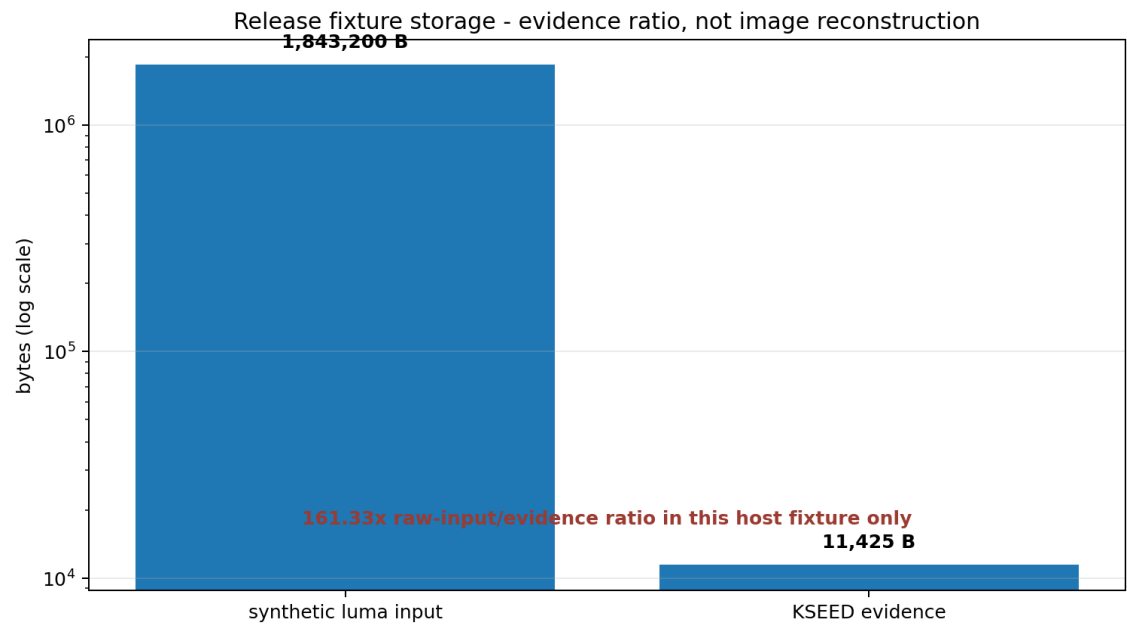


Figure 5. The release fixture ratio compares discarded raw luma bytes with retained evidence; it is not equal-information image compression.

Fixture metric	Measured host result
Analysis frames	120 synthetic frames at 160x96
Raw luma input	1,843,200 bytes
Keyframes / events / voxels	18 / 30 / 1433
KSEED size	11,425 bytes
Nominal ratio	161.33x raw-input bytes / retained evidence bytes
Fixture SHA-256	cad6ef12bb702f8b91bb29608bc6384ed6b5cb9046492735e019313a9c322343
Final SHA chain	c2616adc55927fc400a1f3897138aa6c3d29fabd01a71adebd99a2cbfd451c66

Compression uses delta timestamps/frame indices, quantized inertial values, base-128 varints and Morton key deltas. Zlib level 1 is accepted only when compressed bytes plus a 16-byte margin are smaller than raw payload bytes, so compression never expands a chunk merely to satisfy a policy label.

A finite key, sign, seed or compact state word is not complete geometry. File-size gains remain conditional on the explicit evidence and reconstruction contract.

8. Android and native implementation

Module	Role	Authority
app	Camera2, ImageReader, IMU, UI, thermal tiers, document export, synthesis and presentation	Adaptation
core	Features, matching, SLAM, proposals, verifier, ledger, KSEED writer/reader	Behavioral oracle and map authority
seednative	arm64-v8a JNI bridge	Optional accelerator; no map mutation
native/core	Portable SplitMix64 schedule and CRC32	Host-tested native oracle
tools	Build, bootstrap, source checks, KSEED inspection, PLY conversion, packaging	Independent verification

The Gradle handoff keeps the repaired verified bootstrap. It downloads pinned Gradle 8.13, verifies the required distribution SHA-256 and rejects unsafe ZIP paths before execution. The Android source pins SDK 36, AGP 8.13.2, NDK 29.0.14206865, CMake 3.22.1, Java 17 and arm64-v8a for the native module.

```
./gradlew --bootstrap-self-test
./gradlew :app:clean :app:assembleRelease
adb install -r app/build/outputs/apk/release/app-release.apk
```

Current source counts: 49 Java files, 3 C++ files and approximately 4,642 Java/C++ source lines. The package contains no generated Gradle, build, .cxx, IDE or Python cache tree.

9. Synthetic isolation and Bayer projection

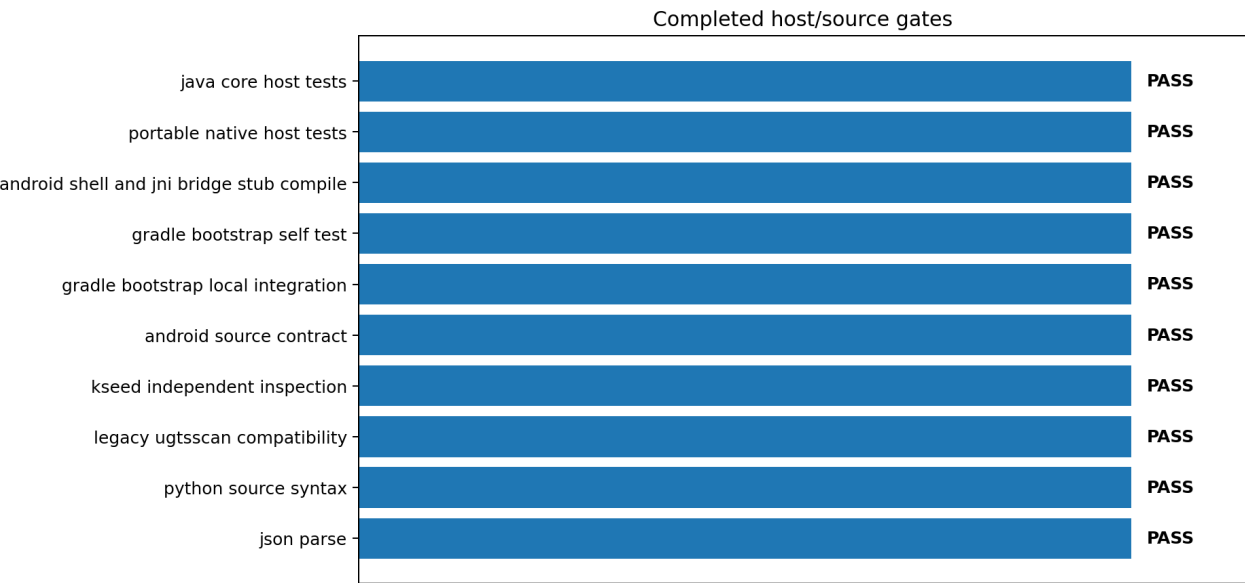
If camera permission or startup fails, the user can run a deterministic 300-frame maximum fixture. It has a fixed seed, generated luma, bounded camera model and synthetic motion. The session flag is set, every proposal carries tag bit 31 and the on-screen banner says that the data is DEMO and not real-world evidence.

Control	Implemented guarantee
Session seed	Fixed deterministic demo seed; exported in KSEED header
Proposal tag	Bit 31 in SpatialProposal tags
Chunk flag	Synthetic fixture flag in KSEED chunks
UI	Visible DEMO banner
Export	Synthetic state flag in final summary
Boundary	Never mixed silently with camera evidence

Bayer4Level implements an exact 8x8 ordered threshold and four output levels 0, 85, 170 and 255. It is a downstream projection utility only. The active scanning view remains the camera preview plus bounded feature/map overlay because the requested application prioritizes tracking accuracy.

Neither Bayer display nor synthetic data can commit authority outside the same proposal verifier.

10. Validation completed



Android assemble and physical POCO measurements remain separate promotion gates.

Figure 6. All completed host and source gates passed.

Gate	Result	Evidence
Java core	PASS	8,465 retained assertions plus 24,369 seed/KSEED assertions
Portable C++	PASS	10,004 schedule/CRC assertions
Android/JNI bridge stub	PASS	Java 17 shell compilation against Android mock surface
Gradle bootstrap	PASS	Self-test plus local fake-distribution hash/extract/execute test
Source contract	PASS	Permissions, dependencies, Camera2, verifier, KSEED, native pins and cache exclusions
KSEED inspector	PASS	Independent Python framing, CRC, zlib, chain and summary verification
Legacy reader	PASS	Previous .ugtsscan sample remains inspectable
Python/JSON	PASS	All tool scripts compile; all JSON parses

A corrupted KSEED byte is rejected by host tests. Deterministic encoding produces identical bytes from the same SessionData. The checked-in fixture's summary stored_bytes equals its actual file length.

11. Android and physical-device promotion gates

Source/core tests	PASS
KSEED independent integrity	PASS
Android SDK/NDK assemble	NOT RUN
POCO install + camera	DEVICE
latency / thermal / battery	MEASURE
SLAM / route accuracy	STUDY

Figure 7. Completed source evidence remains separate from unexecuted phone claims.

The packaging environment did not contain Android SDK Platform 36 plus NDK 29 and did not have a physical POCO X7 Pro connected. Therefore this release does not contain a newly built APK and does not assert camera startup, sustained FPS, memory, thermal, battery, metric error or route correctness.

Mandatory device record	Required observation
Identity	Exact phone model, 12 GB edition, OS/security build, ABI, GPU and display mode
Capture	Camera ID, actual YUV dimensions, actual cadence, intrinsics source and dropped frames
Latency	p50/p95/p99 analysis and KSEED write latency
Resources	Peak Java/native memory, bytes/minute, 5/15/30 minute thermal and battery traces
Integrity	Export/pull, independent KSEED inspection and deterministic replay evidence
Accuracy	Known-distance controls, drift, false/missed events and equal-error calibrated baseline

12. Accuracy, security and evidence boundaries

Topic	What 4.1.1 establishes	What it does not establish
SLAM	Sparse/semi-dense relative visual-inertial reference path	Metric VIO, learned dense depth, bundle adjustment or certified topology re
Metric scale	Known-distance anchor transaction	Drift removal or global calibration
Integrity	Changed bytes, chunk order and chain failures are detectable	Operator identity, truthful place/time, uncompromised hardware or legal cus
Seed	Deterministic schedules, stable IDs and fixtures	Encryption or reconstruction of unstored photons
Compression	Measured retained-evidence byte reduction	Equal-information image compression or universal phone ratio
Native	Portable C++ core compiles and passes host tests	Android ABI load or performance until SDK/device build
Safety	Reason-coded observations and explicit unknown states	Medical, structural, evacuation or legal certification

Stop or simplify the system whenever numeric error or uncertainty exceeds the event margin, synthetic and real evidence cannot be separated, loop closure would require an unimplemented optimizer, KSEED integrity fails, or a conventional calibrated scanner is more accurate or efficient at equal error.

13. Package map and reproducibility

Path	Contents
app/	Camera2/IMU Android shell, UI, thermal control, synthetic demo, KSEED document export
core/	SLAM, verifier, ledger, seed schedule, Bayer utility, KSEED writer/reader and legacy codec
seednative/	Android arm64-v8a library and JNI bridge
native/	Portable C++ seed/CRC implementation and host tests
contracts/	Frame-observation and spatial-proposal schemas
spec/	KSEED binary contract, authority contract and JSON profile
samples/	Deterministic KSEED fixture, inspection summary and retained legacy sample
tools/	Build, bootstrap, host tests, source checks, KSEED inspector, PLY conversion, manifest and release packaging
validation/	Machine-readable validation results and independent KSEED inspection
provenance/ + sources/	Source lineage, substrate alignment and merge boundary
report/	This PDF and reproducible report generator
checksums/ + manifest/	Release SHA-256 inventory and package map

At report-generation time the source tree contains 129 files before final checksums/manifest regeneration. Final release inventory and content-root checks are produced after the PDF is inserted into the package.

```
./tools/generate_kseed_fixture.sh
./tools/run_all_validation.sh
python3 tools/generate_manifest.py
python3 tools/verify_release.py <release.zip>
```

14. Final release definition

UGTS-KC 4.1.1 Atlas Spatial Seed SLAM Fusion is a repaired, compact Android source handoff in which Camera2/IMU observations feed a bounded monocular SLAM front end; stable seed-derived identifiers and measured frame evidence feed an ordered proposal verifier; only accepted proposals mutate keyframe, voxel or metric state; every decision carries reason and hash lineage; and KSEED stores a streamable seed-plus-evidence record with CRC32, conditional zlib and a SHA-256 chain.

RELEASE STATUS. Complete source, host validation, independent KSEED fixture and technical report are delivered. Android assemble, production signing, physical POCO installation and device accuracy/thermal measurements remain the next promotion stage.

Identifier	Value
Version	4.1.1
Codename	Atlas Spatial Seed SLAM Fusion
Target	POCO X7 Pro 12 GB, arm64-v8a
Default storage	KSEED 4.1
Application ID	org.ugts.atlas.slam.pocox7pro
Build pins	SDK 36, AGP 8.13.2, Gradle 8.13, NDK 29.0.14206865, CMake 3.22.1, Java 17
Host validation	PASS
Android APK	Not produced in this environment
Physical device	Not tested in this environment
Merge claim	documented-contract reimplementation; not a byte-identical merge of unavailable 4.1.0 source

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